

ES2631 Critique and Communication of Thinking and Design

AY2025-2026 Semester 2

Assignment 1

Team Oral Presentation on Engineering Conceptual Design Project (30%)

Context

In the early weeks of the semester, through e-lectures, independent and tutorial session activities as well as class discussions, you will be introduced to an engineering reasoning framework that we will use throughout this course. Although there are a number of conceptual frameworks for thinking and reasoning (Elder & Paul, 2020; Ennis, 1996, 2004; Halpern, 2014; Stanovich, Toplak & West, 2008), the **Engineering Reasoning Framework** (Paul, Niewoehner & Elder, 2019) has been specifically adapted for the Engineering discipline. Constant use of the framework helps engineering and design students and professionals cultivate important intellectual traits and dispositions which facilitate critical thought within and beyond the technical domain.

Task

In this first assignment, we place the spotlight on **elements of thought** (Paul, Niewoehner and Elder, 2019) through an exercise that involves identifying a problem within a specific context and proposing an engineering design solution that addresses the problem by taking into consideration the needs and affordances of the context. The intent is not to produce a novel engineering design but to articulate the **reasoning** behind a problem and propose a solution based on the *creative use or adaptation of existing technology*. The proposed solution can be new, significantly improve upon an existing solution (product or process) or modify a solution implemented elsewhere. It should be appropriate for the context. In other words, the focus is on the **thinking and reasoning** undergirding the articulation of a problem and a corresponding possible solution based on an understanding of current approaches as well as context-specific requirements.

This semester, the broad theme for the project is “Sustainable cities and communities” (SDG Goal 11) with a specific focus on sustainable and inclusive **urban transport** within a context (city, community, district or area) **in Southeast Asia**.

General sub-themes within which to situate your research to identify and contextualize the problem are listed below but you may explore other aspects as well. *Narrow your scope* to a specific problem within a category and context/location so that your project is both focused and manageable. For example, if you select road closures and traffic disruptions due to flash floods in Singapore, identifying a specific area such as Bukit Timah Road or Orchard Road that is prone to this problem would allow you to determine what has been done to address the issue, understand the topographical and infrastructural challenges and how the problem can be better managed by improving an existing product/mechanism, system or process.

- Traffic congestion and bottlenecks
- Traffic related noise or air pollution
- Road or rail closures following heavy rains (due to flash floods or landslides)
- Public transport integration (transport and vehicle types such as cars, buses, trains, motorcycles or tuk-tuks), connectivity and disruptions/delays
- Road safety for drivers

- Road safety for pedestrians or cyclists
- Safe and/or affordable transport access for the elderly, women, children or people with disabilities
- Parking shortages and management

Following some preliminary research, identify a context in Southeast Asia (city or area within a city) in which the problem exists. The problem, question or need must be identified via an exploration of existing ways of addressing it and their limitations. In other words, you must show *where* the problem exists, *whom* it affects, *how* or *why* existing approaches are inadequate and *why* it is worth solving. Your proposed solution must be technological, i.e., use principles, theories, models, concepts, frameworks and methodologies specific to engineering. Simply proposing an AI-based solution without a proper understanding of its workings and sufficient consideration of specific mechanisms, concepts, challenges and pitfalls is not acceptable. Likewise, a proposal solely focused on the development of some software such as an App would not qualify as an Engineering solution unless it also has a hardware component.

To facilitate your team discussion and project presentation, we suggest the following steps to take and points to note:

- 1) Review the elements of thought:
 - Question at issue (or problem, in the context of this assignment)
 - Purpose (of this project in relation to the question at issue; based on an understanding of existing approaches to addressing the question within the context and their limitations)
 - Information (facts/data/evidence pertaining to the problem, context and your design solution – these are derived from primary and secondary sources)
 - Inferences (conclusions derived from data and evidence)
 - Concepts (theories and engineering design principles that underlie your solution)
 - Assumptions (ideas that are taken for granted in relation to the problem or your solution)
 - Implications (or consequences of adopting or not adopting your solution)
 - Points of view (different perspectives related to the problem or design of the solution)
- 2) Establish your contextual focus within Southeast Asia and identify a specific problem pertaining to it. Use relevant and credible secondary sources to contextualise the problem.
- 3) Articulate “why” it is a problem, i.e.,
 - What exactly is the gap or concern that needs to be addressed? Whom does the problem affect?
 - What are some existing solutions and why are they not effective? This is important to consider so that your proposed design can improve upon or creatively rethink what has already been tried.
 - What evidence or sources have been interrogated to reach the conclusion that it is a problem (e.g., first-hand observations, gaps or needs in the real world and relevant literature)?

- 4) Provide justifications or make a case for your *engineering or technological* solution, i.e.,
 - What inferences or assumptions have been made?
 - What are the different points of view and how have they been taken into consideration?
 - What concepts, theories or principles of *engineering and/or design* have been applied?
 - What are the potential implications of the proposed solution?
- 5) Sources should be carefully selected and acknowledged in your presentation. Refer to the lecture and handout on **Source Types, Selection, Integration and Attribution** (week 3) for details on how to do so, in order to ensure that your evidence is credible and compelling and to avoid plagiarism. Other than as a search engine, to create graphics for your presentation or as a final proofreading tool, AI in any form **should not** be used for any other aspect of developing and presenting your assignment.
- 6) In your presentation, cover all the relevant and necessary elements as is appropriate but **highlight three elements of thought** for detailed discussion. The team critiquing your presentation in Assignment 2 will focus on these three elements.

Presentation and Submission Guidelines

In groups of four (occasionally three), make an *9-12-minute presentation*. The presentation will be followed by a *7-10 minute Q&A* session in which all team members must participate.

- 1) Each team member must speak for approximately 3 minutes.
- 2) The Q&A is assessed individually and based on both the quality of questions and responses. Team members are urged to take turns asking and answering questions.
- 3) Each presentation and Q&A session are to be recorded. This could be facilitated via a mobile phone, with the recording serving as a reference for Assignment 2, for which you will individually analyse another team's oral presentation. The recording should not be more than 500MB (avoid using 4K recording) and is to be uploaded to MS Teams or any other platform your tutor instructs you to use.
- 4) Your final slides must include a **list of references** in APA format and a **generative AI use declaration**. **If you used AI, please provide details regarding the tool used, prompt and output and how you used the output in the assignment**. The format for the AI declaration is included in the Syllabus and Assessments document (Week 2) as well as the Sources handout (Week 3). Use **in-text citations** on individual slides, where appropriate, to show what your sources are and how you have used them to think through various aspects of the project.
- 5) The presentation slides must be uploaded to the designated Teams folder *before* your presentation and the video recording uploaded on the same day after your presentation.

Assessment criteria

The team project presentation will be assessed on the effectiveness of its rationale and the team's application of the framework. Assessment will be based on the following criteria:

- Critical Thought and Reasoning (40%) [Assessed on a group basis]
 - The presentation explicitly and competently demonstrates understanding of the task requirements:
 - The problem/question and context are well established using appropriate evidence and reasoning.
 - The purpose of the project is clear and relevant.
 - The solution fits the context and is thoughtfully developed; it is based on relevant concepts and supported by facts and data.
 - Inferences, assumptions, points-of-view and implications are appropriate and in keeping with the ERF.
 - The presenters engage the audience effectively and the project arouses audience interest and curiosity.
 - The presenters use relevant and credible sources and other evidence critically and effectively i.e., contextualizing, synthesizing, critiquing, comparing and contrasting, for identification of problem and making a case for solution(s).
 - All sources are credible, timely, relevant and properly cited in APA style throughout the presentation.
 - An AI use declaration is included. If AI was used, details are provided in the recommended format.
- Delivery skills (30%) [Assessed on an individual basis]
 - Few language errors and which do not interfere with the audience's comprehension.
 - Few hesitations and no awkward pauses.
 - Projects confidence and calmness.
 - Clear and accurate pronunciation.
 - Well projected voice with suitable modulation in volume and pitch.
 - Presentation is natural and does not sound scripted – not reading from notes.
 - Upright, open posture with natural and appropriate body movement.
 - Good body position in relation to the projected image.
- Visual aids (20%) [Assessed on a group basis]
 - Successful integration of images/graphics/multimedia and text, demonstrating teamwork.
 - Slides are very well presented in terms of size, alignment, contrast and proximity of components, demonstrating teamwork.
 - Sufficient detail to convey one key point per slide.
 - Excellent transition with presenter leading the slides.
 - Logical progression of slides resulting in a presentation that flows smoothly.
- Q&A (10%) [Assessed on an individual basis]
 - Team members take turns to answer questions and effectively build on one another's responses.
 - Responses are thoughtful, clear and relevant. They further develop, clarify or justify ideas and perspectives discussed during the presentation.
 - Each student also engages other presenting teams by asking relevant questions and/or offering useful suggestions. The questions and suggestions should allow another team to clarify ideas, rethink or build on aspects of their project.

Work Schedule

Please seek approval for your proposed project early so that you will have sufficient time to work on it. Here is a suggested work schedule:

Teaching Week	Activity
Week 2	• Form a group. Give your names to your tutor. • Based on the given theme, identify a problem within a context, understand how it has been addressed and what the limitations are of current approaches; consider possible engineering solutions. • Seek approval from your tutor on your project focus (problem to be addressed and possible solutions). You may email your tutor after the tutorial (and before Week 3) to seek feedback on your ideas,
Weeks 3	• Confirm the topic, problem, context and proposed solution • Develop the engineering solution. • Prepare for the presentation: develop and organize the content and prepare the visual aids.
Week 4	• Attend a group conference with your tutor. • Revise your solution design as well as other components, as appropriate. • Prepare for the presentation: finalise the visual aids and practice.
Week 5	• Assignment 1: Present your project.

References

- Elder, L. & Paul, R. (2020). *Critical Thinking: Tools for Taking Charge of Your Learning and Your Life*. 3rd Edition. Rowman & Littlefield.
- Ennis, Robert H. (1996). *Critical Thinking*. Prentice-Hall.
- (2004). Applying soundness standards to qualified reasoning. *Informal Logic*, 24 (1): 23-39.
- Halpern, D. F. (2014). *Thought and Knowledge: An Introduction to Critical Thinking*. Psychology Press.
- Paul, R., Niewoehner, R. & Elder, L. (2019). *The Thinker's Guide to Engineering Reasoning: Based on Critical Thinking Concepts and Tools* (2nd ed). Rowman & Littlefield.
- United Nations (n.d.) The 17 goals. <https://sdgs.un.org/goals>

West, R. F., Toplak, M. E., & Stanovich, K. E. (2008). Heuristics and biases as measures of critical thinking: Associations with cognitive ability and thinking dispositions. *Journal of Educational Psychology, 100*(4), 930–941.

Appendix

(Paul et al, 2019, pages 14-15)

You may also refer to pp. 31-39 for a more detailed explanation of each element and its application to Engineering research.

Please be selective in your use of the questions below, as they may not all be relevant to your conceptual design project.

Analyzing a Design Using the Elements of Thought

Engineering purpose	What is the purpose of this design?
	What are the market opportunities or mission requirements?
	Who defines market opportunities/mission requirements?
	Who is the customer?
Question at hand	What system/product/process will best satisfy the customer's performance, cost, and schedule requirements?
	How does the customer define "value"?
	Is a new design or new technology required?
	Can an existing design be adapted?
	How important is time-to-market?
Point of view	A design and manufacturing point of view is typically presumed. What other points of view deserve consideration? Stockholders? Component vendors/suppliers? Marketing/sales? Customers? Maintenance/repair/parts? Regulators? Community affairs? Politicians? Environmentalists?
Assumptions	What environmental or operating conditions are assumed?
	What programmatic, financial, market or technical risks have been considered acceptable to date?
	What market/economic/competitive environment is assumed?
	What safety/environmental assumptions are we making? Are these assumptions acceptable?
	What maturity level or maturation timeline is assumed for emerging technologies?
	What happens if we change or discard an assumption?
	What criteria have historically been assumed in defining a "best" or "optimum" solution?
	What assumptions have been made on the availability of materials?
	What manufacturing capability was assumed?
	What workforce skills or attributes have been assumed?

Analyzing a Design Using the Elements of Thought (cont'd)

Engineering information	What is the source of supporting information (handbook, archival literature, experimentation, corporate knowledge, building codes, government regulation)? What information do we lack? How can we get it? Analysis? Simulation? Component testing? Prototypes? What experiments should be conducted? Have we considered all relevant sources? What legacy solutions, shortcomings, or problems should be studied and evaluated? Is the available information sufficient? Do we need more data? What is the best way to collect it? Have analytical or experimental results been confirmed? What insights and experiences can the shop floor provide?
Concepts	What concepts or theories are applicable to this problem? Are there competing models? What emerging theory might provide insight? What available technologies or theories are appropriate? What emerging technologies might soon be applicable?
Inferences	What is the set of viable candidate solutions? Why were other candidate solutions rejected? Is there another way to interpret the information? Is the conclusion practicable and affordable?
Implications	What are some important implications of the data we have gathered? What are the most important market implications of the technology? What are the most important implications of a key technology not maturing on time? How important is after-market sustainability? Is there a path for future design evolution and upgrade? Are there disposal/end-of-service-life issues we need to consider? What are the most important implications of product failure? What design features if changed, profoundly affect other design features? What design features are insensitive to other changes? What potential benefits do by-products offer? Should social reaction and change management issues be addressed?